

"projectile_spawned" Enchantment Component runs predicates and effects on projectiles before all of their data has been assigned

[MC-273281](#) ENCHANTMENTS

Status	Resolved / Fixed
Affected	1.21; 1.21 Release Candidate 1
Fixed in	24w33a
Reported	2024-06-12
Mojira category	Data Packs; Enchantments; Projectiles

REPRODUCTION TARGET

Use version:

1.21

Affected versions (2):

1.21

1.21 Release Candidate 1

Environment:

Minecraft Java Edition 1.21 in a vanilla single-player world. Use Creative mode with commands enabled so the test items and report-supplied enchantment can be applied deterministically. Install the provided projectile_spawned_bug_example_latest.zip datapack in the test world's datapacks directory.

PRECONDITIONS

- A Minecraft Java Edition installation using exactly version 1.21 is available. Do not use fix version 24w33a or any later version for this reproduction.
- The report attachment `projectile_spawned_bug_example_latest.zip` has been downloaded and is accessible to the reproduction environment.
- The Minecraft game directory and the test world's save directory are writable so the datapack can be installed.
- Use the control bound to Use Item, which is right-click by default, when drawing and firing the bow.
- Keep the in-game chat display available because the datapack communicates during each projectile-spawn event.

STEPS TO REPRODUCE

1 REPORT

Start Minecraft Java Edition using an installation configured for version 1.21. Confirm the selected version before creating the test world.

INPUT `Minecraft Java Edition 1.21`

SUCCESS CUE The main menu is open and identifies the running game version as 1.21.

2 INFERRED

Open Singleplayer and create a new world named MC-273281-Repro. Set Game Mode to Creative and enable commands, then create the world.

INPUT `World name: MC-273281-Repro; Game Mode: Creative; Allow Commands: On`

SUCCESS CUE The player has spawned in a new Creative-mode world and can access the hotbar and chat.

3 INFERRED

Wait before modifying the world files. Continue only after the terrain around the player has rendered and movement responds normally.

SUCCESS CUE Nearby terrain is visible, chunks are no longer visibly loading around the player, and player movement is responsive.

4 INFERRED

Pause the game and return to the title screen so the world is closed before installing the datapack.

INPUT `Esc -> Save and Quit to Title`

SUCCESS CUE The Minecraft title screen is displayed and the MC-273281-Repro world is no longer open.

5 REPORT

In the active Minecraft game directory, open the save folder for MC-273281-Repro and then open its datapacks subdirectory. Copy the provided zip attachment directly into that directory without extracting or renaming it.

INPUT

`Copy projectile_spawned_bug_example_latest.zip to <game directory>/saves/MC-273281-Repro/datapacks/`

SUCCESS CUE `projectile_spawned_bug_example_latest.zip` is present directly inside the test world's datapacks directory.

6 INFERRED

Return to Minecraft, open Singleplayer, select MC-273281-Repro, and enter the world again.

INPUT `Singleplayer -> MC-273281-Repro -> Play Selected World`

SUCCESS CUE The player is back in the test world, surrounding terrain is rendered, and movement responds normally.

7 INFERRED

Reload the world's datapacks. Press T to open in-game chat, enter the command with its leading slash, and press Enter.

INPUT `/reload`

SUCCESS CUE The reload completes without reporting that the provided datapack failed to load. The player remains in the world with controls responsive.

8 **INFERRED**

Obtain one bow. Open chat with T, enter the command with its leading slash, and press Enter.

INPUT `/give @s minecraft:bow 1`

SUCCESS CUE A bow is present in the player's inventory or hotbar.

9 **INFERRED**

Obtain arrows for both test shots. Open chat with T, enter the command with its leading slash, and press Enter.

INPUT `/give @s minecraft:arrow 64`

SUCCESS CUE A stack of arrows is present in the player's inventory or hotbar.

10 **INFERRED**

Make the bow the active item. If the bow is in the hotbar—the row of nine item slots at the bottom of the screen—use the mouse wheel or the corresponding number key to select its slot. If it is only in the main inventory, press E, move it to a hotbar slot, close the inventory, and select that slot.

INPUT `Select the hotbar slot containing the bow`

SUCCESS CUE The bow is visible in the player's hand and its hotbar slot is highlighted.

11 **REPORT**

Apply the report-supplied custom enchantment to the held bow. With the bow still selected, press T, enter the exact command from the report including the leading slash, and press Enter.

INPUT `/enchant @s test:arrow_crit_detection`

SUCCESS CUE The command succeeds rather than reporting an unknown enchantment or an inapplicable item, and the selected bow displays an enchanted glint.

12 **INFERRED**

Move outdoors or to another location with a clear firing lane. Face approximately horizontally toward open space, with no block or entity immediately in front of the player. Keep the enchanted bow selected.

INPUT `Use normal movement controls; aim with the mouse`

SUCCESS CUE The enchanted bow remains selected and there is enough open space in front of the crosshair for two arrows to be fired without an immediate point-blank collision.

13 **REPORT**

Fire the first arrow without fully charging the bow. Hold Use Item—right-click by default—for approximately 0.25 to 0.5 seconds, then release while the bow is still partway through its draw animation. Do not hold until the maximum-draw pose.

INPUT `Hold right-click for approximately 0.25–0.5 seconds, then release`

SUCCESS CUE One arrow has been launched from a visibly partial bow draw, and the bow has returned to its resting position in the player's hand.

14 **INFERRED**

Pause briefly between shots without changing items or closing the world. Wait until the first shot has completed its immediate processing and the player can begin drawing the bow again.

INPUT Wait approximately 1 second

SUCCESS CUE The enchanted bow is still selected, is no longer in its draw animation, and the Use Item control can start a new draw normally.

15 **REPORT**

Prepare the fully charged shot. Hold Use Item—right-click by default—and do not release until the bow reaches and remains at its maximum-draw pose. Holding for at least approximately 1.25 seconds avoids releasing before full charge.

INPUT

Press and hold right-click for at least approximately 1.25 seconds; do not release yet

SUCCESS CUE The bow is held at its maximum-draw pose and no longer appears to draw farther while right-click remains held.

16 **REPORT** **TRIGGER**

Release Use Item immediately after confirming that the bow is fully charged, thereby firing the second arrow. Do not perform another attack, command, reload, or item change after releasing.

INPUT Release right-click

SUCCESS CUE The second arrow has just been launched from the fully drawn enchanted bow, after the earlier partially charged arrow.

HANDOFF TO OBSERVATION

The player has fired exactly two arrows from the bow enchanted with `test:arrow_crit_detection`: first from a partial draw and then from a confirmed full draw. The fully charged second arrow has just spawned, chat remains available, and no subsequent command, reload, item change, or additional shot has been performed. Begin observation immediately.

UNCERTAINTIES

- The report attachment's internal files were not supplied in the raw report, so the plan assumes `projectile_spawned_bug_example_latest.zip` is the updated datapack referenced by the report and is usable in final release 1.21.
- The report does not specify single-player versus dedicated server, game mode, world seed, or location. A new single-player Creative world with commands enabled is used as the minimum deterministic setup and is not asserted to be a required bug condition.
- The report describes partial and full charging qualitatively but supplies no exact timing. The timing guidance is an execution aid; the visible bow draw state is the controlling condition.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Trigger reached?	YES / NO
Crash file	
Notes	
Tested by / date	